

EECS 498/CSE 598: Zero-Knowledge Proofs

Winter 2026

Lecture 8

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Agenda for this lecture

- Announcements
- Interactive oracle proofs and polynomial IOPs
- IOP/PIOP security notions
- The Delphian PIOP for R1CS-Sat
- Soundness of Delphian as a PIOP

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Announcements

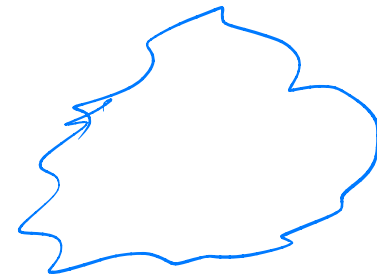
- Project 2 is online
- Autograder is working
- Due date: 2/20

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Interactive oracle proofs

- Mid-80s: IPs Babai, GMR: ZK
Somecheck $IP = PSPACE$ $MIP = NEXP$
- Mid-90s: PCPs $NP = PCP[O(1), \log(n)]$
Proofs checked using randomness
Verifier gets oracle access to proof
- 00s ZKPs from linear PCPs
- 10s: IOPs, P1OPs
1PCP



Interactive oracle proofs

IP

$\forall$

IP with special "oracle"

messages.

IP

sends,

$\forall$

queries

IP

must

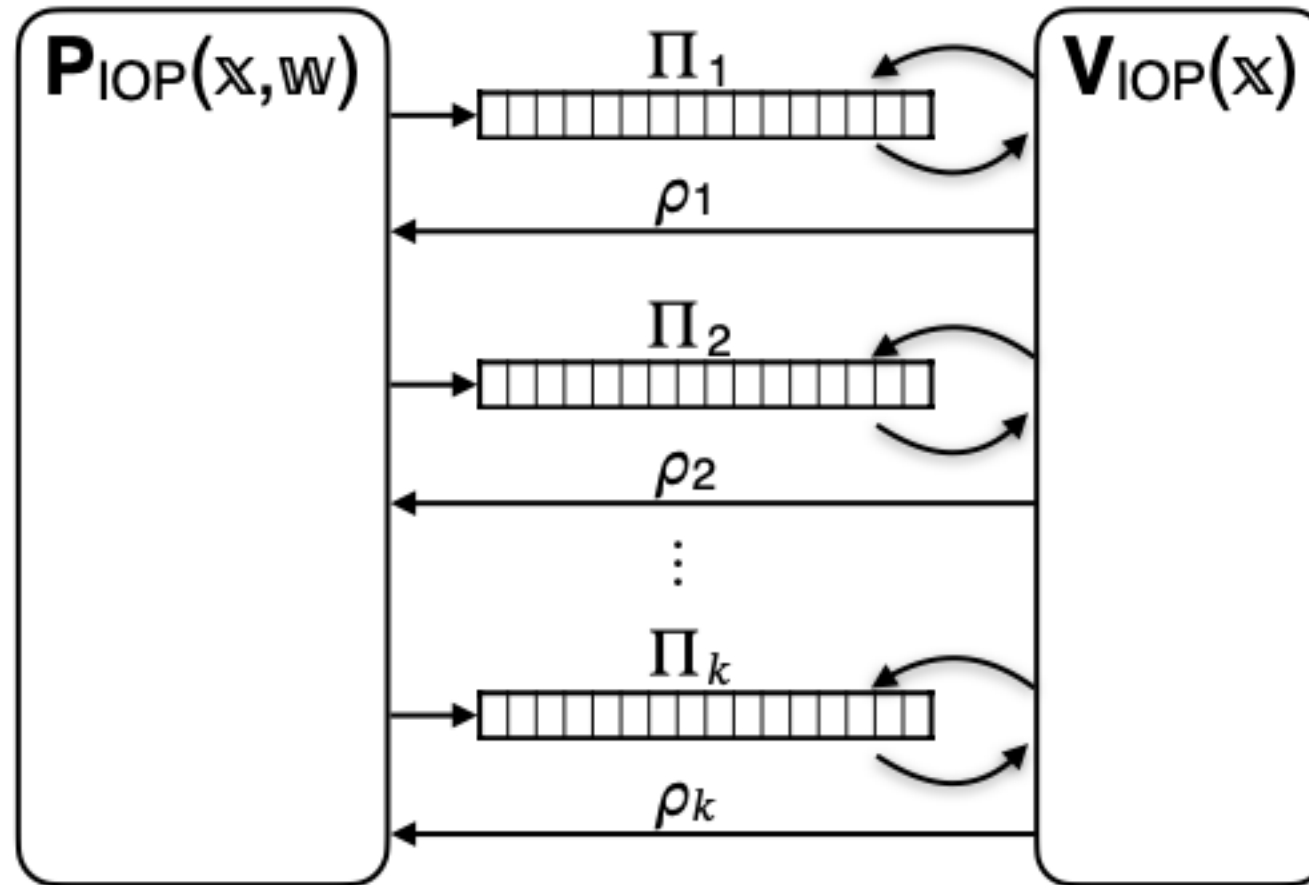
answer

oracle

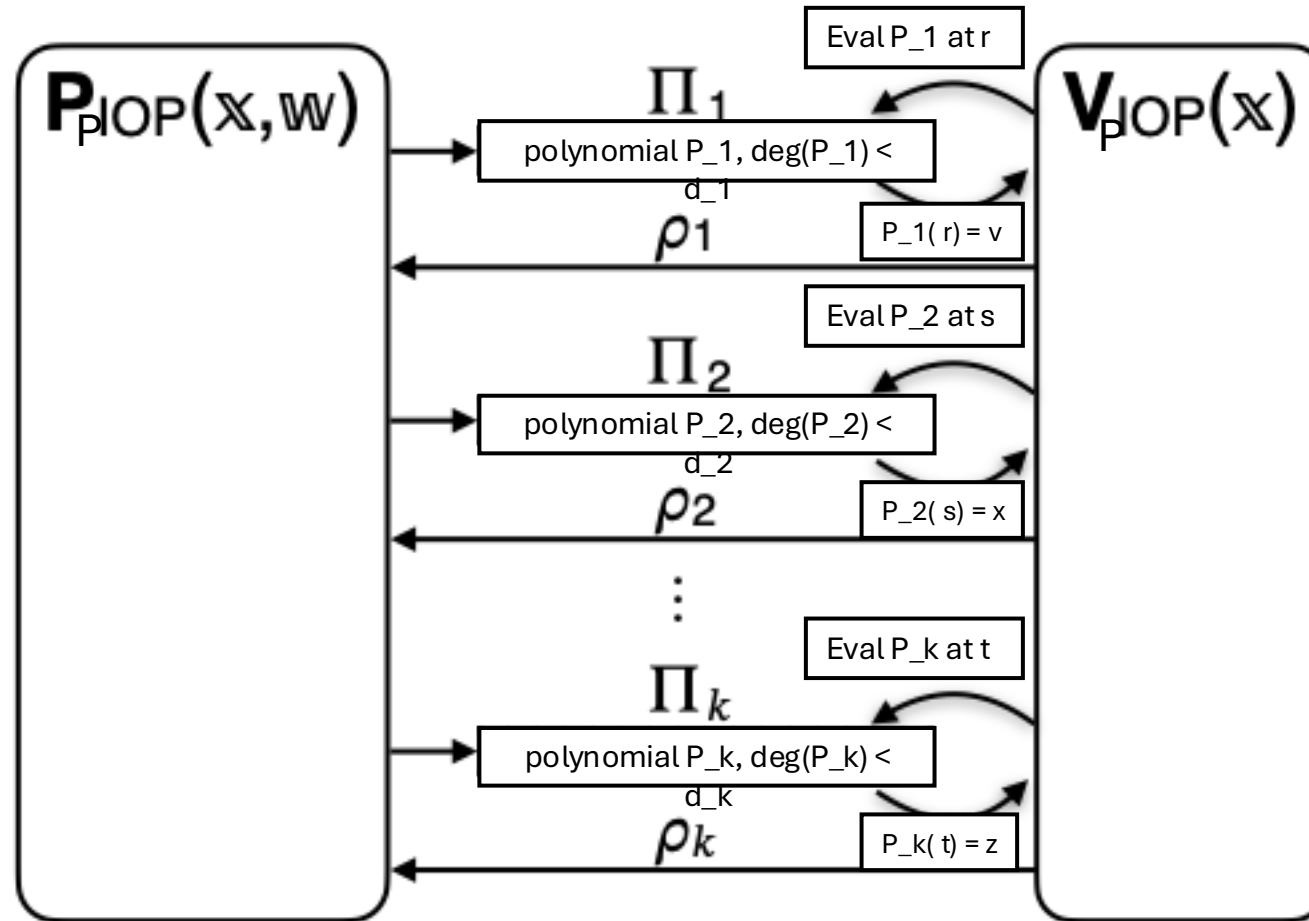
queries

truthfully

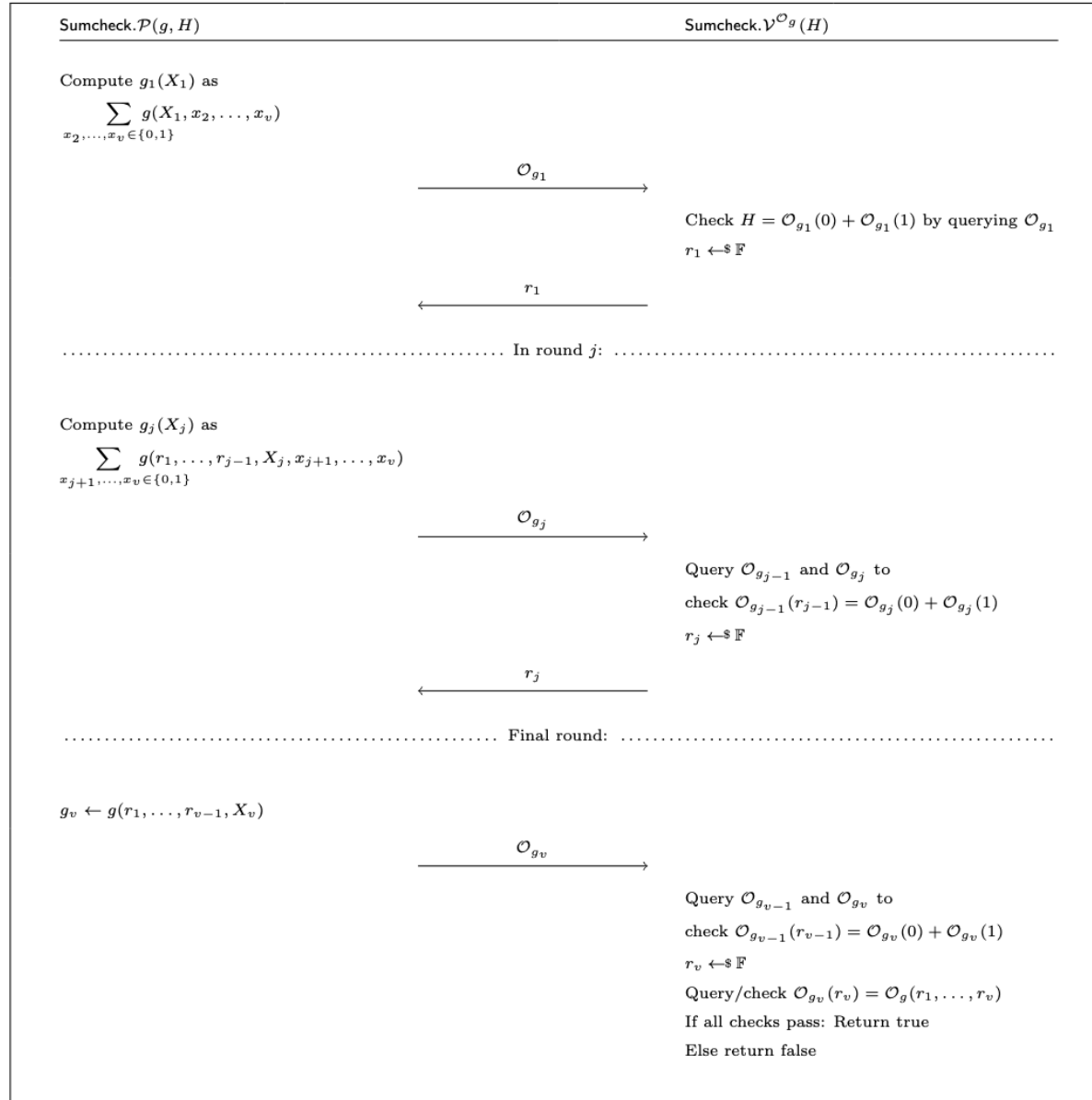
Interactive oracle proofs



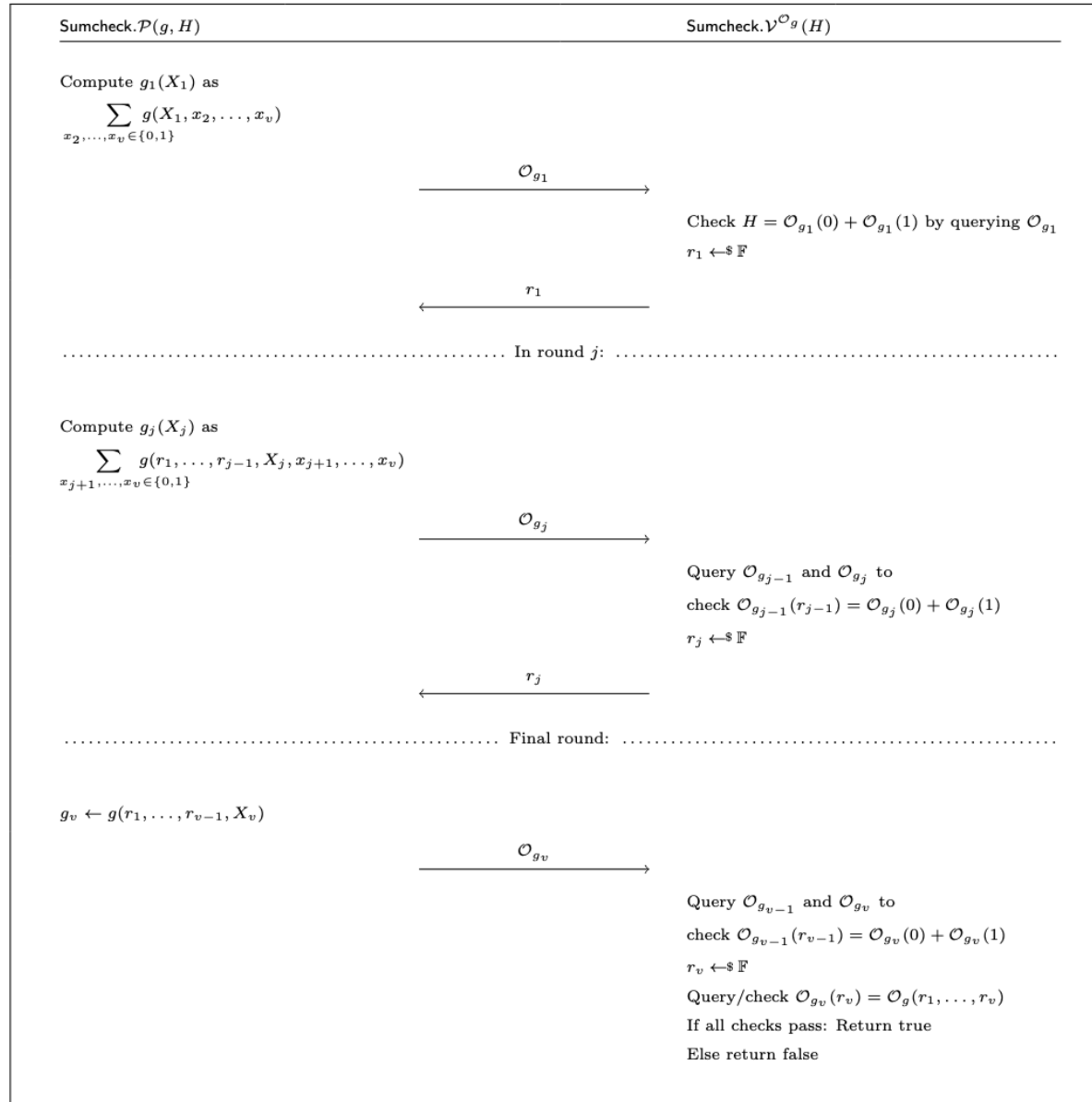
Polynomial IOPs



Sumcheck as a PIOP



Sumcheck as a PIOP



Pop quiz: what is the degree bound for the g_i oracles sent during the proof?

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IOP/PIOP security

Completeness $(X, w) \in \mathcal{R}$

$$\Pr[\text{out}_{\mathcal{V}}[P(X, w)] \leftrightarrow \mathbb{1}[X(X)] = 1] = 1$$

ϵ -Soundness: for $X \notin \mathcal{L}$, P^*

$$\Pr[\text{out}_{\mathcal{V}}[P^*(X)] \leftrightarrow \mathbb{1}[X(X)] = 1] \leq \epsilon$$

Round-by-round soundness

non-adaptive soundness

Knowledge soundness

State-restoration soundness

witness-extended emulation

Simulation soundness

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$$A, B, C: \mathbb{F}^{n \times n}$$

$$x, w$$

Delphian PIOP

(first phase)

For RICS-Sat
 $\hookrightarrow$ can prove
 all of NP

$$x \in R_{A,B,C} \text{ if } \exists w:$$

$$z = (x || w)$$

$$Az \circ Bz = Cz$$

$$\mathbb{P}_{\text{Delphian}}(A, B, C, x, w)$$

$$\forall (A, B, C, x)$$

$$F_{\text{res}}(x) : \left(\sum A(x, j) z(j) \right) \left(\sum B(x, j) z(j) \right) - \left(\sum C(x, j) z(j) \right)$$

$$F_{\text{res}} \in [0, 13^{\log n}] \rightarrow \mathbb{F}$$

$$F_{\text{res}} \text{ is } 0 \text{ on } [0, 13^{\log n}] \text{ iff } z \text{ satisfies } A, B, C$$

Delphian PIOP

(first phase)

$$g_{res}(x) = \left(\sum \tilde{A}(x, j) \tilde{Z}(j) \right) \left(\sum \tilde{B}(x, j) \tilde{Z}(j) \right) - \left(\sum \tilde{C}(x, j) \tilde{Z}(j) \right)$$

$$g_{res} : \mathbb{F}^{\log n} \rightarrow \mathbb{F}$$

$$g_{res}(x) = F_{res}(x) \quad \text{on} \quad \{0, 1\}^{\log n}$$

$\forall x, g_{res}(x) = 0$ on hypercube iff $\exists$ statistics

Let $\tilde{eq}$ be MLE of $eq : \{0, 1\}^{\log n} \times \{0, 1\}^{\log n} \rightarrow \{0, 1\}$

Delphian PIOP

(first phase)

Let $\tilde{e}_q$ be MLE of $e_q: \{0,1\}^{\log n} \times \{0,1\}^{\log n}$

Thm: If $\text{gres}(x) \neq 0$ for some $x \in \{0,1\}^{\log n}$, $\rightarrow \{0,1\}^{\log n}$

$$\Pr_{\chi \leftarrow \mathbb{F}^{\log n}} [\sum \tilde{e}_q(\chi, x) \text{gres}(x) = 0] \leq \frac{1}{|\mathbb{F}|}$$

Delphian PIOP

(first phase)

$\mathcal{P}(A, B, C, x, w)$

$\mathcal{V}(A, B, C, x)$

compute multilinear extension $\tilde{w}$ of w

$\mathcal{O}_{\tilde{w}}$

r

compute $h(x) = \tilde{e}\tilde{q}(r, x)g_{\text{res}}(x)$

$\mathcal{O}_h$

$\text{SCPIOP}[\mathcal{P}_{sc}(h, 0) \leftrightarrow \mathcal{V}_{sc}^{\mathcal{O}_h}(0)]$
 $\mathcal{V}_{sc} \rightarrow r' = (r'_1, \dots, r'_{\log n}), H'$

for $M \in \{A, B, C\}$,

$p_M(x) = \tilde{f}_M(r', x)\tilde{z}(x)$

$v_M = \sum_{x \in \{0,1\}^{\log n}} p_M(x)$

$v_A, v_B, v_C, \mathcal{O}_{PA}, \mathcal{O}_{PB}, \mathcal{O}_{PC}$

$r \leftarrow \$ \mathbb{F}^{\log n}$

$$h(x) = \tilde{e}\tilde{q}(r, x)g_{\text{res}}(x)$$

$$h(r'_1, \dots, r'_{\log n}) = \tilde{e}\tilde{q}(r, \vec{r}')g_{\text{res}}(\vec{r}')$$

$e \leftarrow \tilde{e}\tilde{q}(r, r') ; G' \leftarrow H'/e$

[Note G' is claimed to be $g_{\text{res}}(r')$]

$$g_{\text{res}}(\vec{r}') = \left(\sum \tilde{A}(\vec{r}', j) \tilde{z}(j) \right) \cdot \left(\sum \tilde{B}(\vec{r}', j) \tilde{z}(j) \right)$$

Check $v_A * v_B - v_C = G'$

$$= \left(\sum \tilde{C}(\vec{r}', j) \tilde{z}(j) \right)$$

Delphian PIOP

(second phase)

$$P_M(r''_M) = \tilde{M}(r', r''_M) \cdot \tilde{Z}(r''_M)$$

..... Complete for $M \in \{A, B, C\}$:

$$\begin{array}{c} \text{SCPIOP}[\mathcal{P}_{sc}(p_M, v_M) \leftrightarrow \mathcal{V}_{sc}^{\mathcal{O}^{PM}}(v_M)] \\ \xleftrightarrow{\hspace{1.5cm}} \\ \mathcal{V}_{sc} \rightarrow r''_M = (r''_{M,1}, \dots, r''_{M, \log n}), H'' \\ \\ \xleftrightarrow[\mathcal{V} \text{ gets } w_M \leftarrow \tilde{w}(r''_M[: -1])]{\text{Query } \mathcal{O}_{\tilde{w}}(r''_M[: -1])} \end{array}$$

$$\begin{aligned} z_M &= (1 - r''_M[-1])\tilde{x}(r''_M[: -1]) + r''_M[-1]w_M \\ e_M &= \tilde{f}_M(r', r''_M)z_M \\ \text{Check } H'' &= e_M \end{aligned}$$

..... Final checks

If any subprotocols output $\perp$ or
checks fail, output $\perp$, else True

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Soundness of the Delphian PIOP

Theorem 2. *Let $n = 2^\ell$ and $A, B, C \in \mathbb{F}^{n \times n}$ be an R1CS over $\mathbb{F}$ and take $x \in \mathbb{F}^{n/2} \notin \mathcal{L}_{A,B,C,n,\mathbb{F}}$. Let $\mathbb{V}_D$ be the Delphian PIOP verifier. For any malicious PIOP prover $\mathbb{P}^*$,*

$$\Pr[\text{out}_{\mathbb{V}_D} [\mathbb{P}^*(A, B, C, n, x) \leftrightarrow \mathbb{V}_D(A, B, C, x)] = 1] \leq \frac{9\ell}{|\mathbb{F}|}$$